

Glow the Buzz: a VR Puzzle Adventure Game Mainly Played Through Haptic Feedback

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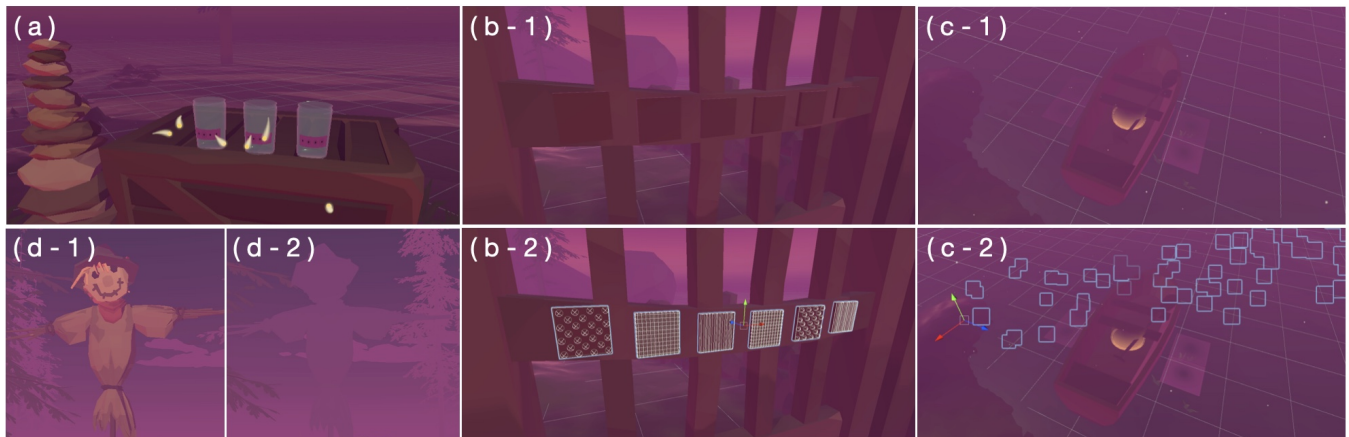


Figure 1: In-Game Images of *Glow the Buzz*. (a) Haptic rhythm puzzle; players confront six visually identical bugs with different haptic rhythm patterns. (b) Haptic texturing puzzle; 1, all patches are visually identical. 2, the visualizations of patches' colliders. (c) Haptic direction puzzle; 1, when players are on the boat, vibrations mimicking wind transfer to the haptic vest. 2, the visualizations of particles. (d) World activation; the objects regain their colors when players touch them. 1, activation state. 2, deactivation state.

ABSTRACT

Virtual Reality (VR) has become a popular tool, leading to increased demands for various immersive VR games for players. In addition, haptic technology is gaining attention as it adds a sense of touch to the visual and auditory dominant Human-Computer Interface (HCI) in terms of providing more extended VR experiences. However, most games, including VR, use haptics as a supplement while mostly depending on the visual elements as their main mode of transferring information. It is because the complexity of haptic in accurately capturing and replicating touch is still in its infancy. To further investigate the potential of haptics, we propose to *Glow the Buzz*, a VR game in which haptic feedback serves as a core element using wearable haptic devices. Our research explores whether haptic stimuli can be a primary form of interaction by conceiving

iterative playtests for three puzzle designs - rhythm, texture, and direction. By proposing a VR haptic puzzle game that cannot be played without haptics, the study concludes that haptic technology in VR has the potential extendability. The study also suggests elements that enhance discriminability of haptic stimuli in each puzzle.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI); • Software and its engineering → Interactive games.

KEYWORDS

Virtual Reality, VR games, Haptic-central Gameplay, Haptic Texturing, Haptic Rhythm, Haptic Direction

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1 INTRODUCTION

The increasing popularity of VR has led to a need to diversify game content in order to maintain user engagement. Similar to traditional video games, most VR games focus on auditory and visual elements. However, there are promising advantages when haptic sensations - a technology that stimulates the touch and tactile experience [15][19] - are added to the game. Previous researches have shown that incorporating tactile stimuli can enhance the user's immersion in the content [11][8][1], provide new and exciting ways to interact with the game environment in real-time [2][19], and enable the user to manipulate virtual objects [5].

According to the study conducted by Suzuki et al., tactile stimuli itself has a potential to induce the player's behavior throughout gameplay [23]. However, not many VR games that actively utilize haptic stimulants as the primary means for its play have been developed. Most VR game contents that take advantage of haptic effects are racing games [22], rhythm games [6], and adventure games [21], which use tactual stimuli as an additional feature. Here, we introduce *Glow the Buzz*, a haptic-based VR puzzle adventure game in which tactual stimuli serve as a core element, driving the game's narrative. We aim to extend the range of tactual experience and its possible applications by classifying haptic experience into three categories - rhythm, texture, and direction. Furthermore, we used state-of-the-art wearable haptic devices to explore the possibility of amplifying the tactile sensation in VR games as a bodily experience. Our research questions focus on understanding the following:

- **RQ1:** How can haptics be designed to make players clearly distinguish the stimuli of haptic rhythm, haptic texture, and haptic direction?
- **RQ2:** Can haptic stimuli be the primary form of interaction in gameplay?

2 RELATED WORK

2.1 Effects of Tactile Sensation

There have been numerous studies on utilizing haptic technology to enrich user experience in VR games. For instance, by combining audiovisual sense and tactile sensation, the level of immersion experienced by the player can be increased [8]. Konishi et al. explored a high level of immersive embodied experience through the vibrotactile feedback of the Synesthesia Suit [14]. In HaptiCast, players can experience dynamic movement of virtual objects through the haptic device [2]. Additionally, although there are attempts for incorporate tactile feedback with devices, they are primarily used to enhance or simulate reality experiences in conjunction with audiovisual feedback [20][10]. Hence, we discovered a design opportunity to further explore the tactile sensation itself, and how it can be implemented in games as a main component.

2.2 Related Games

2.2.1 Games using Haptic Feedback. The game contents that actively use tactile feedback could find in the field for rehabilitation and educational purposes targeting individuals with disabilities [18][12][17][25][4]. These studies align with our work in terms of incorporating tactual feedback in the game. However, these types

of games may not be suitable for general audiences since they were designed based on specific purposes and target groups.

2.2.2 Haptic-based Games. Tactile-centered VR game [23][9][7] was a rare form among the existing haptic games, almost focused on utilizing tactual sense as an additional feature. These works demonstrate the potential of haptic feedback to enhance the sense of presence in VR and to provide crucial information in problem-solving games. However, it is still unclear whether players engaged in games that mainly used haptic stimuli because audiovisual stimuli were also given together. To develop rich haptic-based game content, we aim to delve into different categories of haptic perception and utilize them as a key element that leads the game's story.

3 INTERFACE INNOVATION

Game design in *Glow the Buzz* focuses on incorporating haptic stimuli as a primary element in gameplay. To develop a game that progresses through tactile sensation itself, we designed three in-game puzzles, each implementing three different types of haptics - rhythm, texture, and direction. Thus, understanding design factors to enable players to differentiate the haptic stimuli in each puzzle was important.

In the game, the concept of using various types of tactual feedback based on the research that classified different categories of haptics was introduced. There have been studies on various categories of haptic perceptions such as frequency, amplitude, moving pattern, texture, position, direction, and rhythm [3][13][24][16][26]. For example, Martinez et al. used vibrotactile gloves to discriminate specific shapes and texture patterns [16]. Also, many researchers utilized vibrotactile technology for navigation tasks [3][13][24]. However, previous VR games using haptic feedback [23][7] utilized it in a more passive and unified way, as a vibrating notification to the player. Thus, we aim to further contribute to these emerging areas of research by focusing on the method of implementing diverse tactile perception that would naturally expand the haptic experience in gameplay in VR games.

Additionally, the three different haptic stimuli are experienced through tactile gloves and a haptic vest. To maximize the tactile experience, in *Glow the Buzz*, neither auditory nor visual cues are given. Since the player needs to solve the puzzle only focusing on the sense of touch, all three haptic designs are improved through iterative playtests. Thus, regarding the design of haptic rhythm, we considered various factors such as vibration intensities and duration. The position and the timing of the player's hands toward the object were also studied when developing a distinguishable haptic texture design. Lastly, for the haptic direction, the player can experience getting hit by a gust of wind through a haptic vest. Here, whether to exaggerate the stimuli by adding nonrealistic feedback was also one of the main considerations.

4 GAME CONCEPT AND PLAY

4.1 Game Concept

Glow the buzz is a VR puzzle game that places three puzzles that can only be solved by getting clues from haptic feedback from interactions between the player and the puzzle related in-game

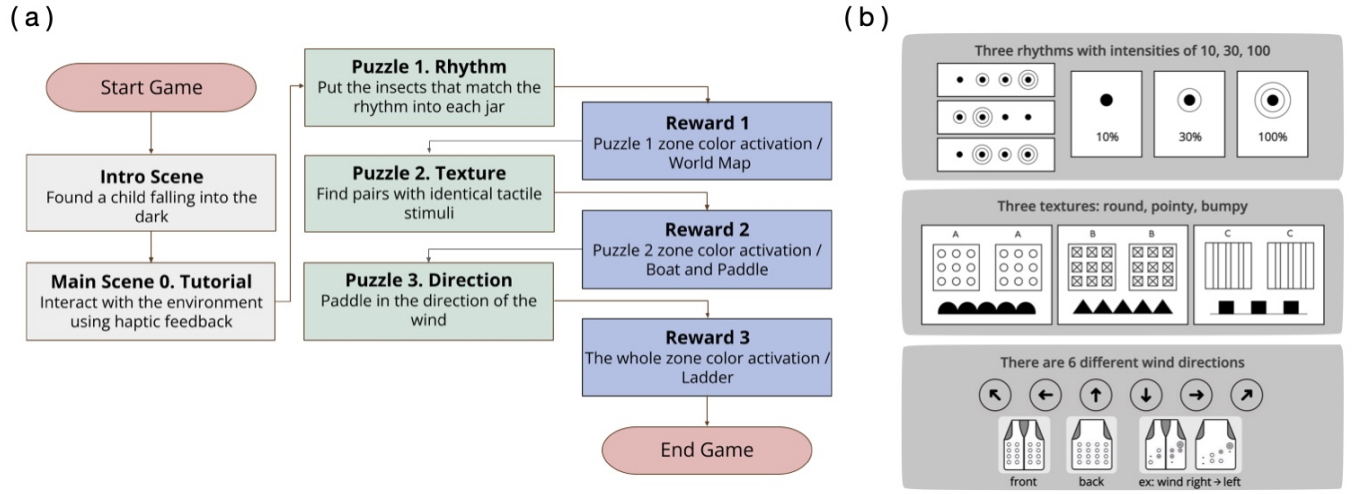


Figure 2: (a) Game Design Flowchart of *Glow the Buzz*. (b) (top) Rhythms and intensities of haptic feedback for Puzzle 1. (middle) Textures of haptic feedback for Puzzle 2. (bottom) Wind directions of haptic feedback for Puzzle 3.

objects. To optimize the haptic design for each puzzle and the user experience, three playtests with a digital prototype were conducted.

4.2 Technical Note

Glow the buzz was developed using Unity 2021 game engine and requires Meta Quest 2, haptic gloves, and a vest made by bHaptics Inc. Each glove is equipped with six motors: 5 for each finger and 1 for the wrist, and the vest comes with 20 motors in a 5 rows, 4 columns configuration on both the front and the back. Controllable parameters for the haptic actuators are the intensity, duration, and time interval between each vibration. Additionally, the game is built with a hand-tracking feature. The hand-tracking feature shows good performance with bare hands but becomes unstable with haptic gloves on. To remediate this effect, blue latex gloves were worn on top of the haptic gloves.

4.3 Game Design

After the intro scene, the game begins in an environment with minimal visual stimuli intended to make the players concentrate more on haptics. The game is divided into four levels. The first level is the tutorial. The player can instantly see a window providing information on the locomotion method, world-activation, and hint system using a non-playable character called Bandi¹. The other three levels are made up of haptic puzzles described below. After solving one puzzle, teleportation zones to the next puzzle are activated. World Activation is an extra feature that players can experience during gameplay. The virtual objects vibrate when players touch them, giving a sense of interaction with nature (See Figure 1. d.) The objects are initially darkened and unsaturated but reveal their actual color upon activation.

4.3.1 Iterative Process. Three iterative playtests were conducted to assess the effectiveness of puzzle designs during the game's

development. The haptic stimuli discriminability in each puzzle was examined with the question, "Were you able to recognize the differences between haptic stimuli in the puzzles?" with a 5-point scale in every playtest. Haptic rhythm and texture puzzles were tested in all playtests, whereas the haptic direction puzzle was only included in the third, along with a direction classification test. Based on insights from the playtests, the puzzles were modified to enhance the players' ability to distinguish between haptic stimuli accurately. The significant changes made in puzzle designs during each playtest are explained below.

4.3.2 Puzzle 1: Rhythm. Puzzle 1 is designed to identify different rhythms of vibration (See Figure 3. a.) One cycle of rhythm consists of four vibrations with the same duration but with a combination of three different intensities. The rhythms come from six visually identical bugs to haptic gloves when the player holds them (See Figure 1. a.) The player is given three glass jars, each with a drawing representing one of six rhythms. The puzzle is solved when the player puts the bugs in all three jars that match the drawings on them. If the wrong bug is put in, then it escapes from the jar after three seconds.

In the first playtest ($n = 7$), the rhythms consisted of four intensity levels: 10%, 30%, 70%, and 100%. and were three seconds long. However, during the playtests, players found it difficult to remember the rhythms with such length and were unable to sort out mid-level intensities (30% and 70%).

In the second playtest ($n = 4$), intensity levels were reduced from four to three, removing 70% level. This is because it was more difficult to distinguish 70% and 100% than to distinguish 10% and 30%. The duration was also shortened from 3 sec to 1.8 sec. As a result, most players were able to distinguish the differences in each insect's rhythm.

In the third playtest ($n = 10$), there was no significant change.

Result. Participants in each playtest rated their ability to distinguish between each insect's haptic stimuli on a 5-point scale (1 =

¹A name taken from the Korean term for firefly

not at all; 5 = very much). A one-way ANOVA test was performed to compare the discriminability of the haptic rhythm design across all three playtests. It was revealed that there were significant differences in distinguishing the haptic stimuli used in the rhythm puzzle between each playtest [$F(2, 18) = 11.55, p < 0.01$.] Tukey's HSD Test for multiple comparisons found that the mean value of first and third tests was significantly different ($p = 0.001$).

4.3.3 Puzzle 2: Texture. Puzzle 2 involves finding matching pairs among six visually identical patches, where players use haptic feedback to recognize textures (See Figure 1. b.) The haptic design for each texture, such as pointy, bumpy, and round shapes, was controlled by adjusting the intensity and duration of vibration (See Figure 3. b.) The designed haptic was sent to the gloves at the moment of touch and when a player rubs the patches. When the player's hand stayed on the patches, weak stimuli were given as to replicate the sense of continuous touch. When the player simultaneously touches two patches with the same haptic texturing, the edges of the pair are instantly highlighted to confirm that the player found the matching pair. Finding all three pairs would unlock the gate to the next area.

In the first playtest, the colliders that make players feel the type of textures were arranged on the patches (See Figure 1. b-2.) They emitted a designed vibration pattern to the player's haptic gloves whenever hand colliders touched the gloves. Even when the hand collider stayed on the patch collider for a longer time, the same designed pattern was played. The players mentioned that it was difficult to distinguish between patches because of low-vibration clarification.

In the second playtest, there was no significant change.

In the third playtest, the design of haptic pattern was changed depending on the timing, whether the hand collider starts to hit with the patch collider (touch-enter) or stays on it (touch-stay). The vibration pattern is released according to each pattern's design during touch-enter. In contrast, weak and continuous stimuli are released during touch-stay, replicating the sense of touch in the real world. Although playtesters could distinguish between the patches relatively better than before, some still required clarification. This is because the combination of stimuli from the touch-enter and touch-stay varies depending on the speed at which the patch is swept.

Result. Same survey and ANOVA test were performed as Puzzle 1. There were no significant differences ($p = 0.304$).

4.3.4 Puzzle 3: Direction. Puzzle 3 involves players navigating a boat by rotating it against the direction of the wind, implemented with a particle system in Unity. (See Figure 1. c.) The avatar's body has 20 cubes in a 5-by-4 configuration on both the front and the back. When particles collide with a particular cube on the avatar's body, they create a haptic effect on a motor with the same position on the vest the player is wearing. To progress through the puzzle, players must feel and identify the direction of particle flow among six directions and align the boat accordingly at each zone. Once the boat points the wind direction, it moves on to the next zone.

In the first and second playtest, Puzzle 3 was not included.

In the third playtest, particle systems were deployed in each zone to spray particles in one direction. Playtesters found the puzzle challenging due to irregular and unclear feedback from the particle

system. An additional direction classification test was conducted using two versions of haptic direction patterns designed with the "bHaptics player" to address this issue. One version, *normal*, provided normal wind directions to prevent vibration in body parts that the wind cannot reach. The other version, *go-through*, added vibrations on the side of the body that does not feel the wind to mimic the feeling of wind going through the body. Playtesters were asked to select one of the four options for each of the eight questions in each version. Playtesters performed better in the go-through version, which provided additional stimulation to mimic the sensation of wind going through the body.

Result. A paired samples t-test was conducted to compare players' performance in distinguishing the haptic wind direction implemented by haptic stimuli in each *normal* and *go-through* version. The results indicated a significant difference between the *normal* version ($M = 0.85, SD = 0.15$) and the *go-through* version ($M = 0.63, SD = 0.18$) [$t(9) = -6.19, p < 0.001$].

5 PLAYER EXPERIENCE EVALUATION

5.1 Procedures

The player experience was evaluated only in the third playtest. A total of 10 participants were recruited. Most participants had prior experience with VR devices and one player had previous experience with a game that utilizes haptic feedback. The playtest consisted of three stages. First, participants took a haptic direction classification test with a haptic vest worn. Once the test was finished, participants were asked to put on haptic gloves and a VR HMD to play the game. Researchers provided brief instructions and allowed participants to play the game without a time limit. The overall playtime varied between 20-30 minutes. After the game, participants were asked to fill out questionnaires to evaluate their experience. The questionnaires included the SUS Presence questionnaire and the IMI for subscales, which measure the participant's level of interest/enjoyment, effort/importance, and pressure/tension related to the game. Participants were then asked to provide feedbacks on the perceived difficulty of distinguishing different haptic stimuli in each puzzle.

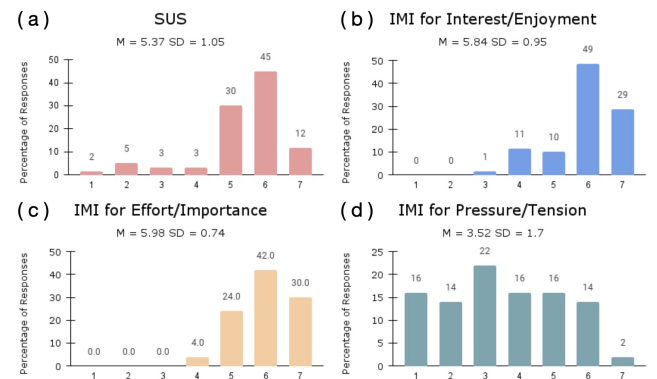


Figure 3: (A) SUS (B) IMI for Interest/Enjoyment (C) IMI for Effort/Importance (D) IMI for Pressure/Tension

5.2 Results

5.2.1 Questionnaires. Based on a 7-point Likert scale, participants' SUS score, or sense of presence in VR, marked the average score as 5.37 with SD at 1.05 (See Figure 4.) IMI scores with subscales of interest/enjoyment, effort/importance, and pressure/tension were also measured to rate the game's entertainment value. The result in interest/enjoyment ($M = 5.84$, $SD = 0.95$) showed participants enjoyed playing *Glow the Buzz* and a low score on pressure/tension ($M = 3.52$, $SD = 1.7$) implied that the game did not give too much pressure when solving the puzzles. On the other hand, effort/importance score was even higher than interest/enjoyment ($M = 5.98$, $SD = 0.74$), suggesting several features or bugs in the game were challenging to the players, such as unstable hand tracking.

5.2.2 Interviews. The perspective on haptic-based gameplay with minimized visual stimuli was overall positive. All the participants unanimously agreed that using haptic feedback as the primary means of gameplay made the game more immersive and even more interesting than the games that use haptic feedback as a supplementary tool to the gameplay. Players especially felt the use of haptics for navigation tasks was novel. However, one player mentioned that the haptic devices were uncomfortable to wear. The views on world activation were positive as well. Of the participants, 70% agreed that it was an exciting feature and were inclined to touch the objects to feel haptics or to see the change of color.

6 DISCUSSION

In this study, two research questions were investigated, focusing on the identification of design factors of haptic rhythm, texture, and direction for distinguishing tactual information (**RQ1**) and examining the potential of haptic feedback in a broader range of gameplay scenarios (**RQ2**). In addition, iterative playtesting was used to investigate users' ability to distinguish diverse tactual information and examine haptic stimulation's effectiveness in solving puzzles.

RQ1. Several critical design factors were identified. In *haptic rhythms*, the length of the vibration should be considered to match a person's memory capacity, and a number of vibration intensities that constitute a rhythm should not exceed three. Moreover, the threshold for distinguishing a given tactile stimulus should be taken into account. Regarding *haptic texturing*, the timing and intensity of vibration given to the hand are essential factors. However, low haptic resolution of gloves used in the study limited the implementation of fine haptic texturing. When haptic feedback was utilized for specific objectives like finding *haptic directions*, incorporating virtual and exaggerated stimuli was found to be more effective in differentiating them than realistic stimuli.

RQ2. The study found that using haptic feedback to solve puzzles was naturally accepted by players. Use of haptics successfully provided more realistic sense of touch when grabbing or touching as well as sufficient immersion, demonstrating the feasibility of haptic feedback as the primary form of interaction in games. Finally, survey and interviews measuring game experience resulted in high scores indicating the acceptability of haptic feedback.

7 LIMITATION AND FUTURE WORK

Several limitations were encountered during the playtests. For instance, replicating textures for Puzzle 2 using only six motors with few controllable parameters was particularly cumbersome. The use of motors capable of producing finer vibration and higher haptic resolution with more parameters may enhance the user experience. To address the unstable hand tracking issue, blue latex gloves were worn over haptic gloves, which remediated the issue, but players still felt uncomfortable when grabbing or touching objects. According to the interviews, it tended to break players' immersion.

In future work, the result of the haptic direction classification test needs to be reflected in Puzzle 3 and additional haptic puzzles may incorporate new categories of tactile perception, such as frequency, amplitude, and moving patterns. Multiple types of haptic stimuli could also be combined to create more complex puzzles.

8 CONCLUSION

Glow the Buzz has proved the feasibility of the use of haptics as the central role of gameplay. The survey shows that participants were satisfied with the game, while it also contributed to changing their previous perception of haptics in an optimistic way. In order to enrich tactual experiences, haptic puzzles representing rhythm, texture, and direction were designed by an iterative process. In each puzzle, players focused only on the feedback sent to the hands, the chest, and the back via haptic gloves and a vest. The puzzles within the game gained positive feedback, extending the possibility of tactile content in VR games.

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